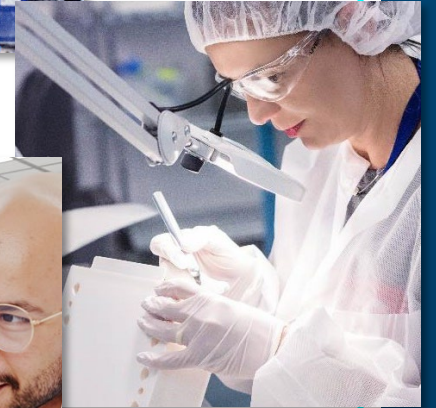


“Shaken not stirred”, why James Bond was right: the next generation PAT solution for downstream processing

Karol Łacki, James Peyser, Alan Chan, Jay West





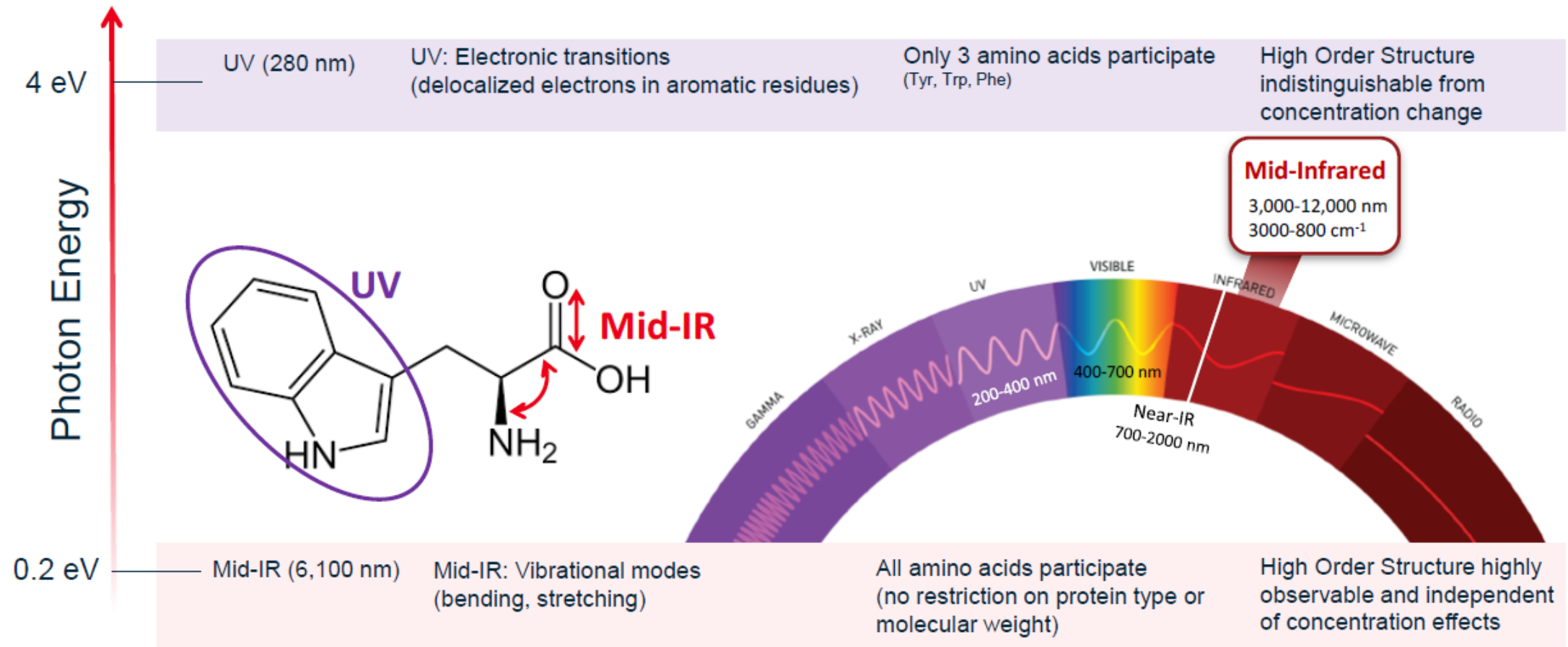
vibrate /vaɪˈbreɪt/

to shake slightly and quickly, or to cause something to do this, in a way that is felt rather than seen or heard



Short intro to mid range infrared spectroscopy

UV vs. mid-IR for quantification



Courtesy of Daylight Solutions

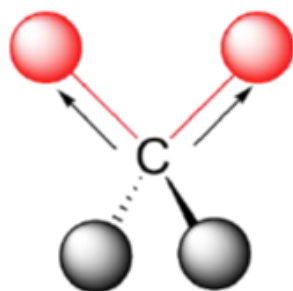
Vibrational modes of molecules

Examples

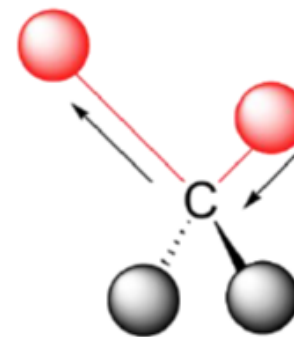
Goal:

Precisely measure the amount of light lost at each frequency due to the coupling of light to the vibrational modes of the molecule.

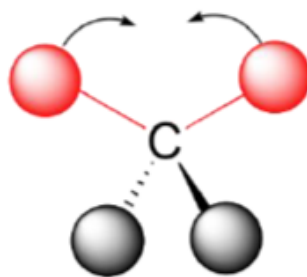
Every molecule will have a set of unique vibrational frequencies – the frequency position and strength of the absorption depend on the local chemical environment of the molecule.



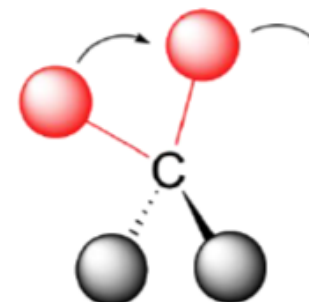
symmetric stretching



asymmetric stretching



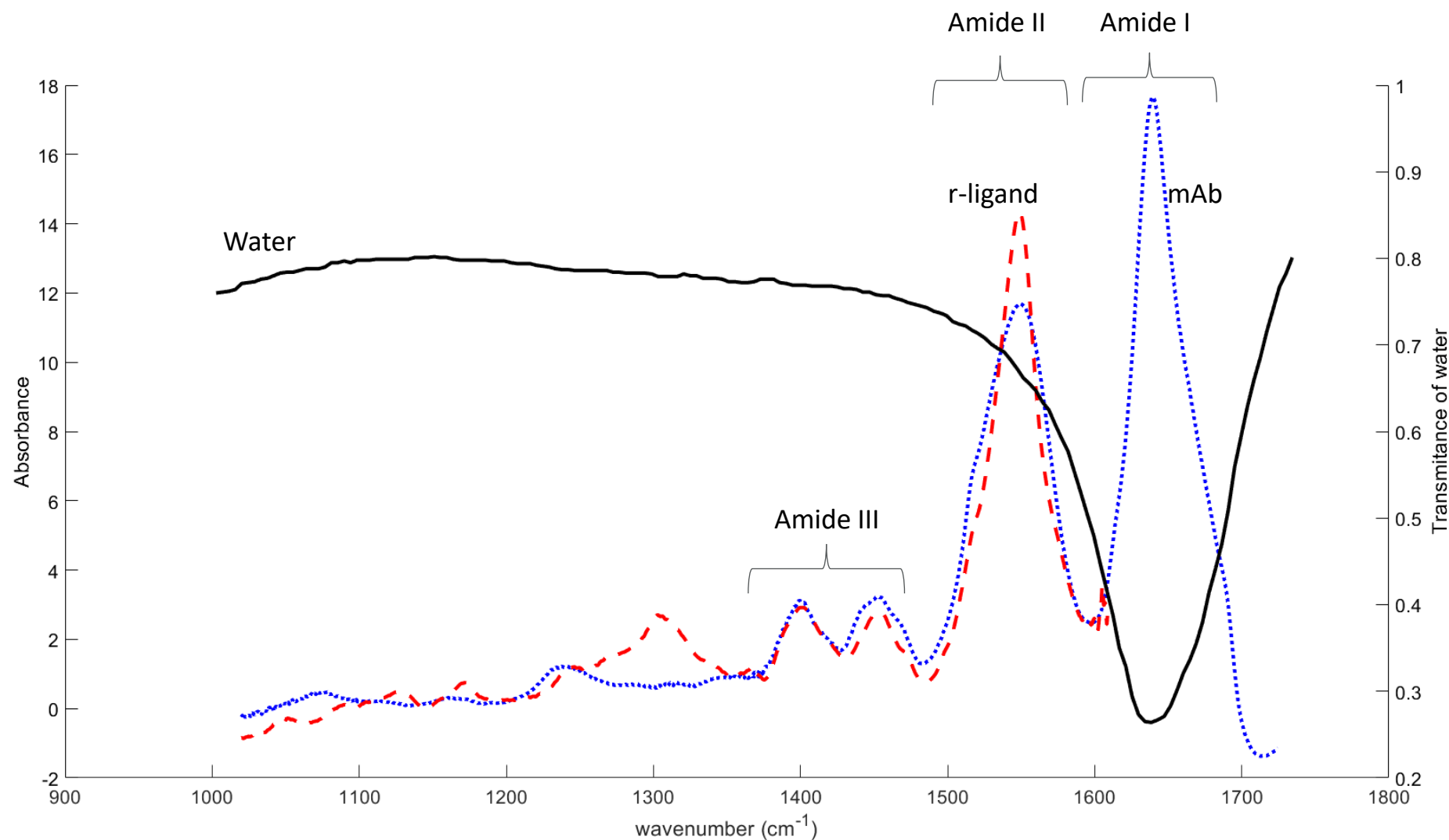
scissoring



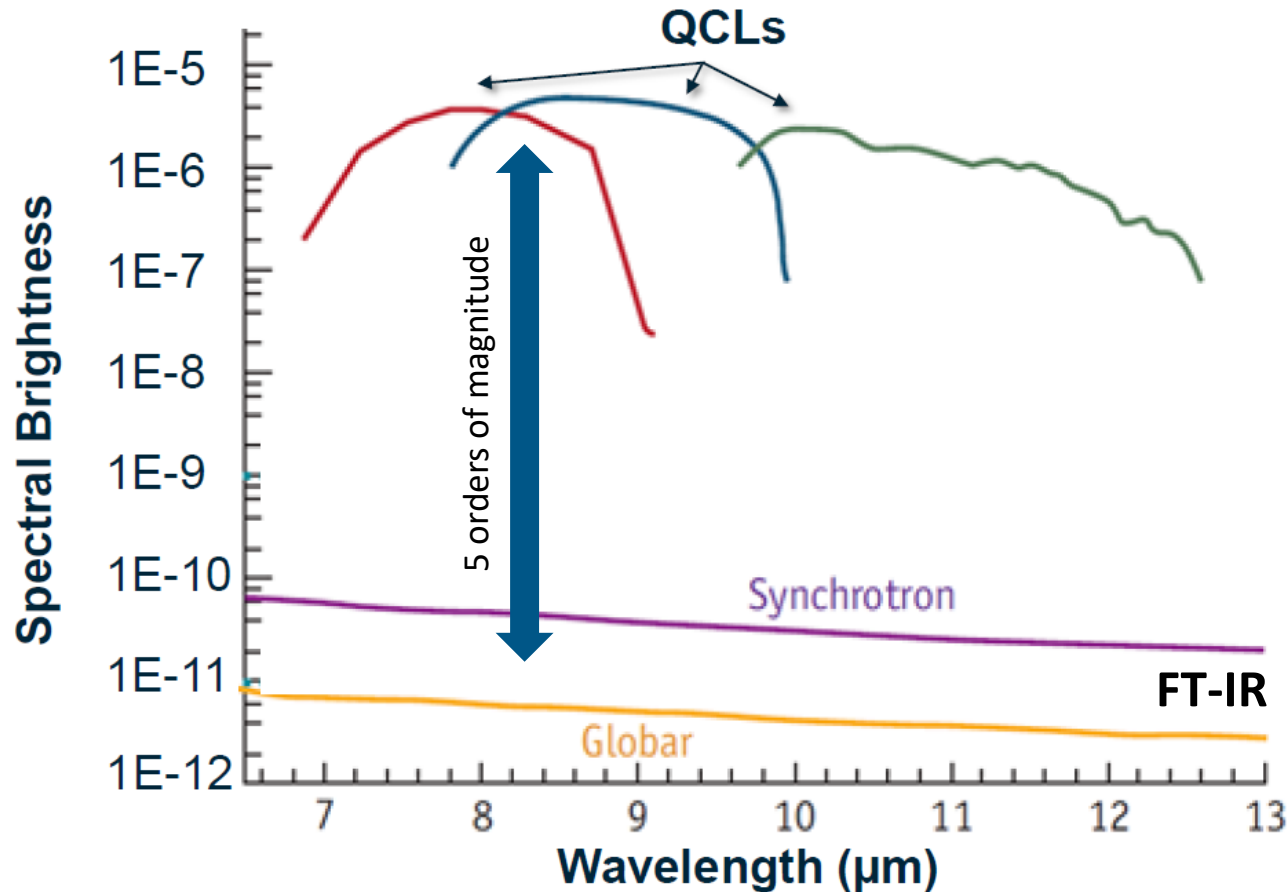
rocking

Vibrational mid IR Spectroscopy

Structural and Quantitative Material Information



Advantages of Quantum Cascade Lasers (QCL)



QCLs = High-power + Broad-tuning

- Higher sensitivity
- Real-time update
- Superior beam control
- No cryogenic cooling

More powerful laser leads to shorter exposure times and, hence, faster measurements

Repligen mid-IR QCL PAT Instrument design

Culpeo® QCL-IR Liquid Analyzer

- Originally developed by DRS Daylight Solutions
- Licensed by Repligen in late 2022 for use in Bioprocessing
- Uses **Quantum Cascade Laser** to perform direct transmission measurements – 2 lasers available with different wavenumber coverage
 - **Laser-Sugar (LA-S) covers 1000-1400 cm^{-1}**
 - Sugars, buffers, DNA and RNA, amino acids, protein, glycoproteins, and detergents
 - **Laser-Protein (LA-P) covers 1350-1750 cm^{-1}**
 - Protein secondary structure, aggregation
 - **Three types of flow cells**
 - 19 μm (LAP), 51 μm (LAP/LAS) or 76 μm (LAS)
- In-line real-time data
 - Rapid analytical method: down to 0.1 second to collect a spectrum
 - High sensitivity/accuracy/low limit of detection



PAT potential of mid-IR spectroscopy

Prior studies focusing on use of mid-IR for monitoring of downstream processes

Examples

Chromatography

“In-column ATR-FTIR spectroscopy to monitor affinity chromatography purification of monoclonal antibodies”; Boulet-Audet, M., Kazarian, S. & Byrne, B.. Sci Rep 6, 30526 (**2016**).

“In-line Fourier-transform infrared spectroscopy as a versatile process analytical technology for preparative protein chromatography”, S. Großhans, M. Rüdte, A. Sanden, N. Brestrich, J. Morgenstern, S. Heissler, J. Hubbuch; Journal of Chromatography A, 1547 (**2018**) 37–44

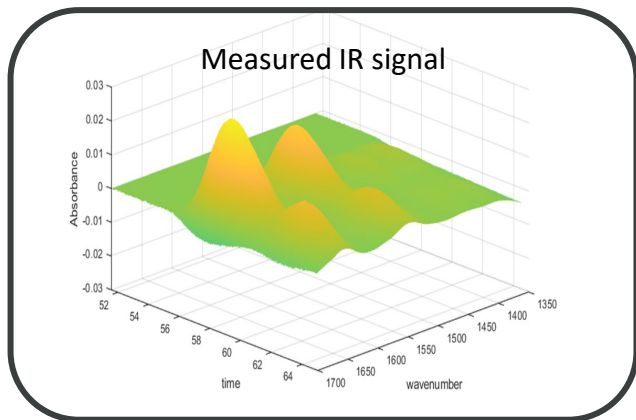
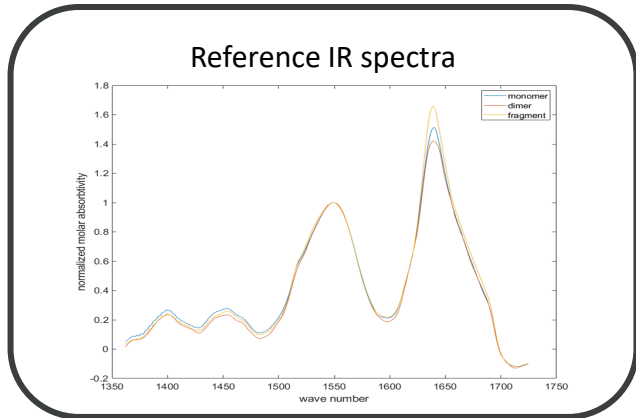
“QCL-IR Spectroscopy for In-Line Monitoring of Proteins from Preparative Ion-Exchange Chromatography Christopher” K. Akhgar, J. Ebner, O. Spadiut, A. Schwaighofer and B. Lendl; *Anal. Chem.* **2022**, 94, 5583–5590;

UF/DF

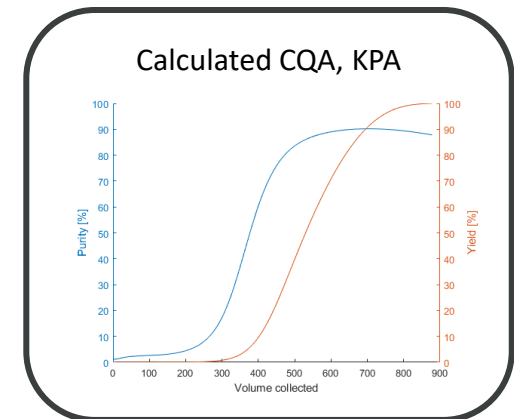
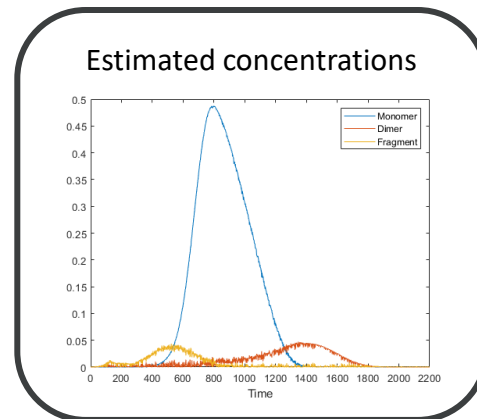
“Real-time monitoring of quality attributes by in-line Fourier transform infrared spectroscopic sensors at ultrafiltration and diafiltration of bioprocess” DP Wasalathanthri, H Feroz, N Puri, J Hung, G Lane, *et. al.*; *Biotechnology and Bioengineering* 117 (12), 3766-3774, **2020**.

“In-line monitoring of protein concentration with MIR spectroscopy during UFDF”, A. Milewska, G. Baekelandt, S. Bouteieb, V. Mozin, A. Falconbridge; *Engineering Life Sciences*. **2022**; e2200050.

mid-IR based PAT solution for bioprocessing



mid-IR technique is compatible with any of the monitoring principles:
at-line, on-line, in-line, off-line



A single mid-IR detector for multi-attribute measurements:
e.g., product concentration; product and process related impurities concentrations; conductivity monitoring, and even pH

Math behind mid-IR

Beers-Lambert law

$$A_{i,j} = \epsilon_{i,j} * C_j * l$$

$$A_{\text{tot},i} = \sum_{j=1}^N A_{i,j} = \sum_{j=1}^N C_j * \epsilon_{i,j} * l$$

where: $A_{i,j}$ is the absorbance of compound j at wavenumber i , C_j is the concentration of compound j , $\epsilon_{i,j}$ is the molar absorptivity coefficient of compound j at wavenumber i , and l is the pathlength

To estimate concentration of each of the N components comprising the analyzed sample, a system of N linear equations needs to be solved

$$\begin{cases} C_1 \epsilon_{1,1} l + C_2 \epsilon_{2,1} l + C_3 \epsilon_{3,1} l + C_4 \epsilon_{4,1} l + \dots C_N \epsilon_{N,1} l = A_{\text{tot},1} \\ C_1 \epsilon_{1,2} l + C_2 \epsilon_{2,2} l + C_3 \epsilon_{3,2} l + C_4 \epsilon_{4,2} l + \dots C_N \epsilon_{N,2} l = A_{\text{tot},2} \\ C_1 \epsilon_{1,3} l + C_2 \epsilon_{2,3} l + C_3 \epsilon_{3,3} l + C_4 \epsilon_{4,3} l + \dots C_N \epsilon_{N,3} l = A_{\text{tot},3} \\ \vdots \\ C_1 \epsilon_{1,N} l + C_2 \epsilon_{2,N} l + C_3 \epsilon_{3,N} l + C_4 \epsilon_{4,N} l + \dots C_N \epsilon_{N,N} l = A_{\text{tot},N} \end{cases}$$

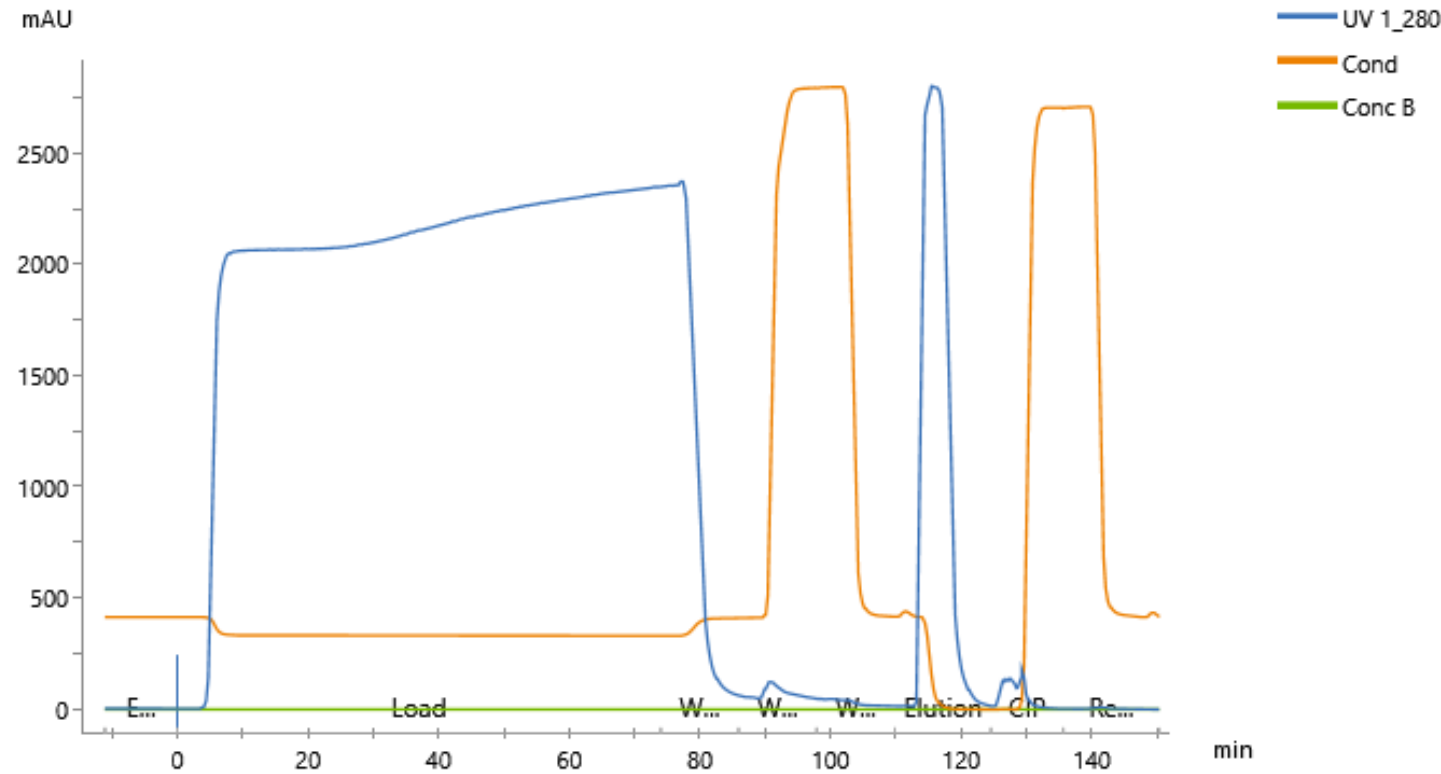
Note: If concentrations of N components are known, then the same system of equations can be solved to calculate the individual components molar absorptivity coefficients at each wavenumber at which the absorbance values are recorded.

Examples

Example #1

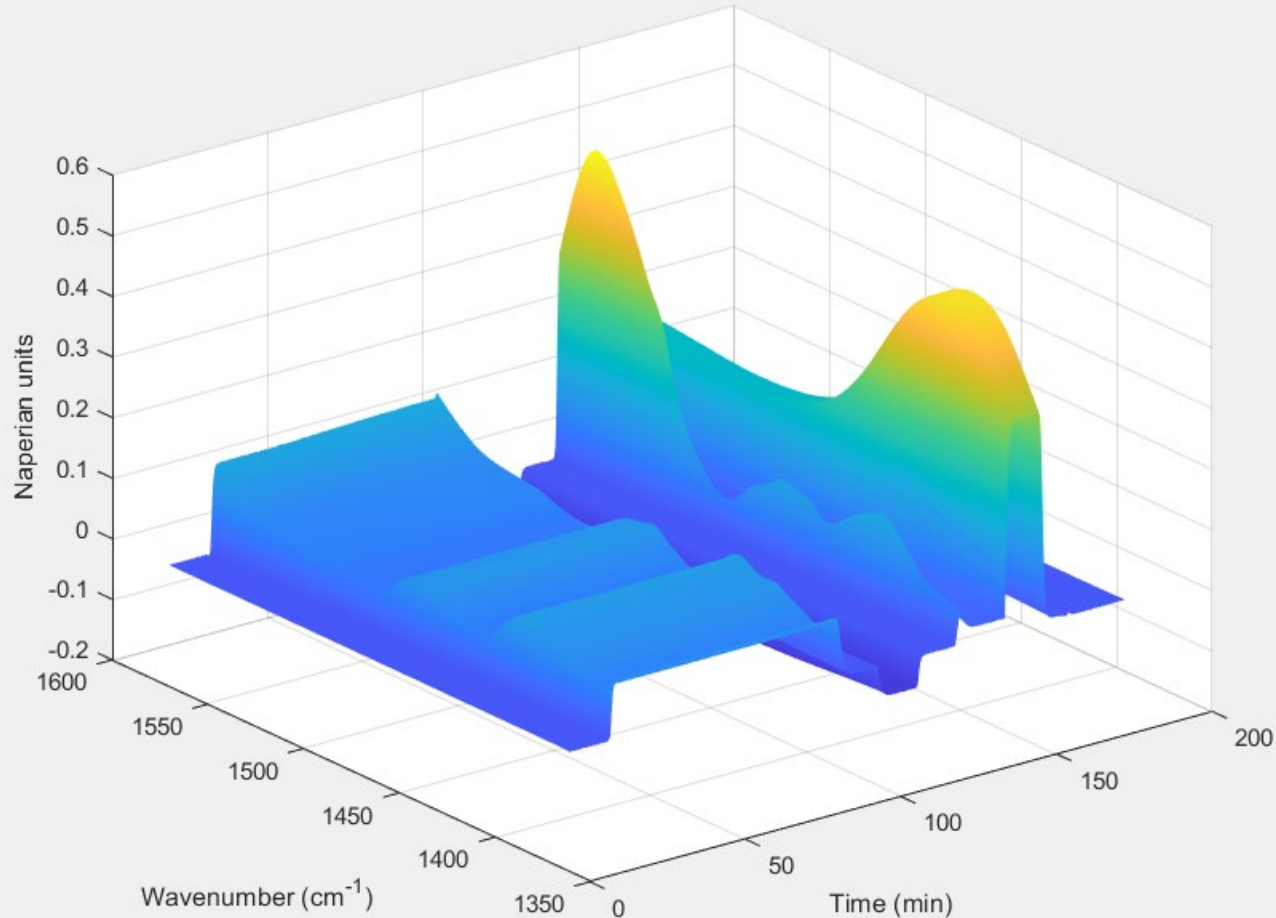
Monitoring of a capture step

Equilibration	B1	@ 2 min RT
Load	mAb + HCP	2g/L @ 2 min RT
Wash 1	B1	@ 2 min RT
Wash 2	B2	@ 2 min RT
Wash 3	B1	@ 2 min RT
Elution	E1	@ 2 min RT
CIP	CIP	@ 2 min RT
Re-Equilibration	B1	@ 2 min RT



Example #1

Monitoring of a capture step

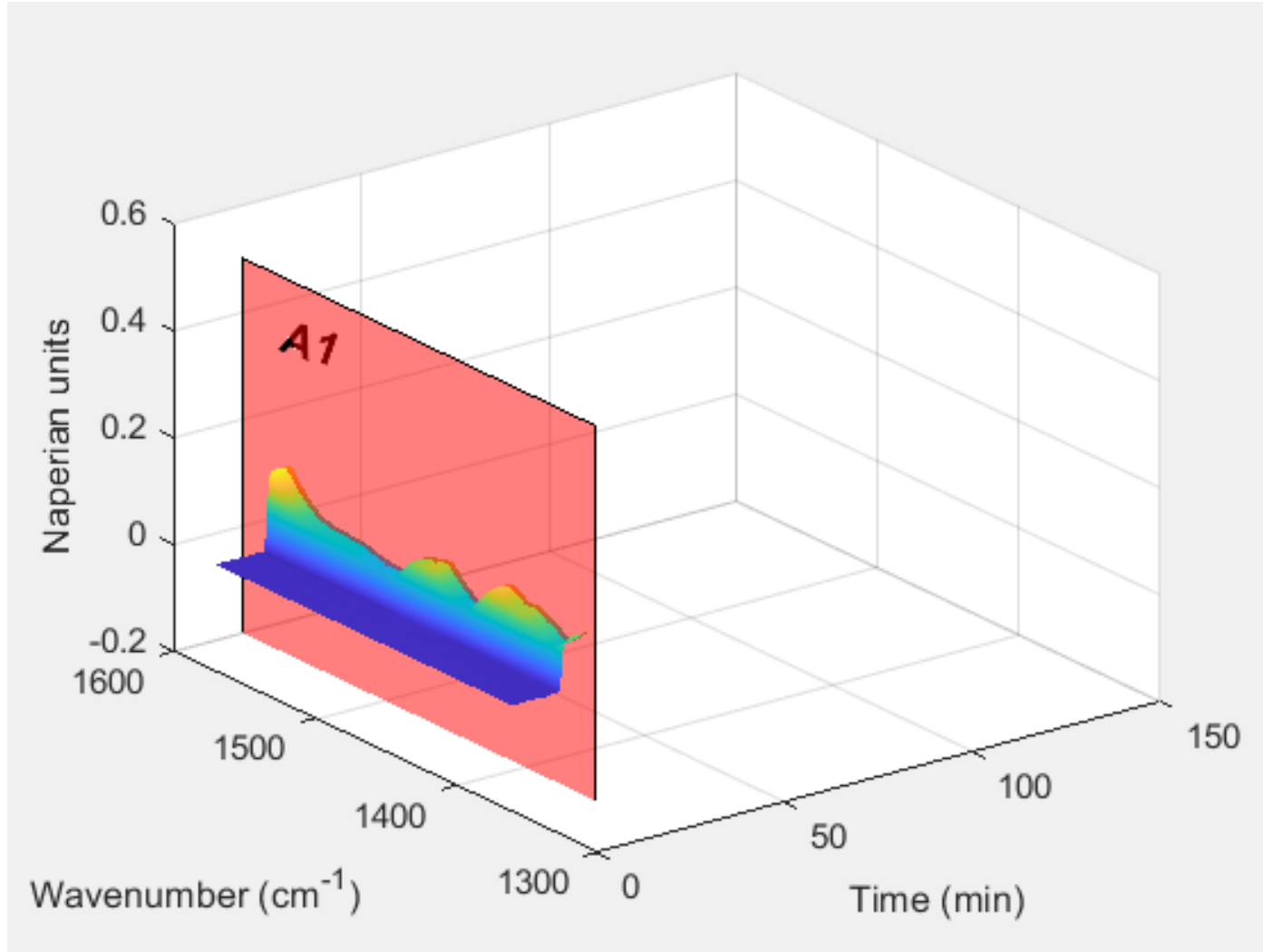


Two-dimensional array of data. At each time point, one can take a slice through the data and extract the characteristic mid-IR spectra that gives a unique fingerprint of the column eluent composition at that time point

The spectra can be extracted for a chosen range of wavenumbers that may be more information reach from a given column eluent composition perspective

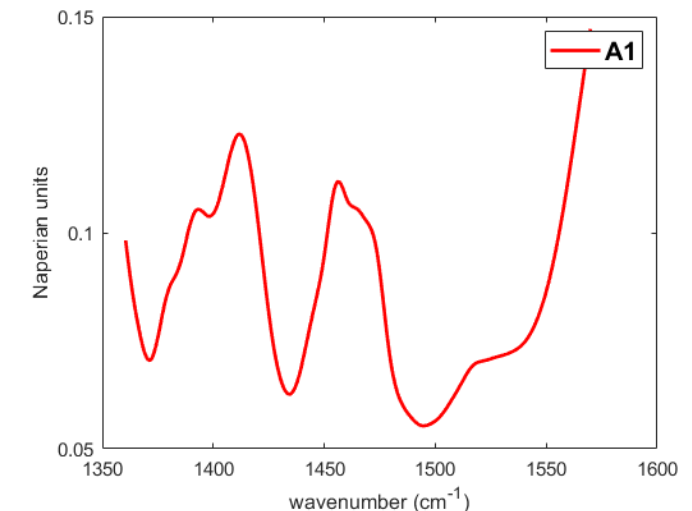
Example #1

Monitoring of a capture step



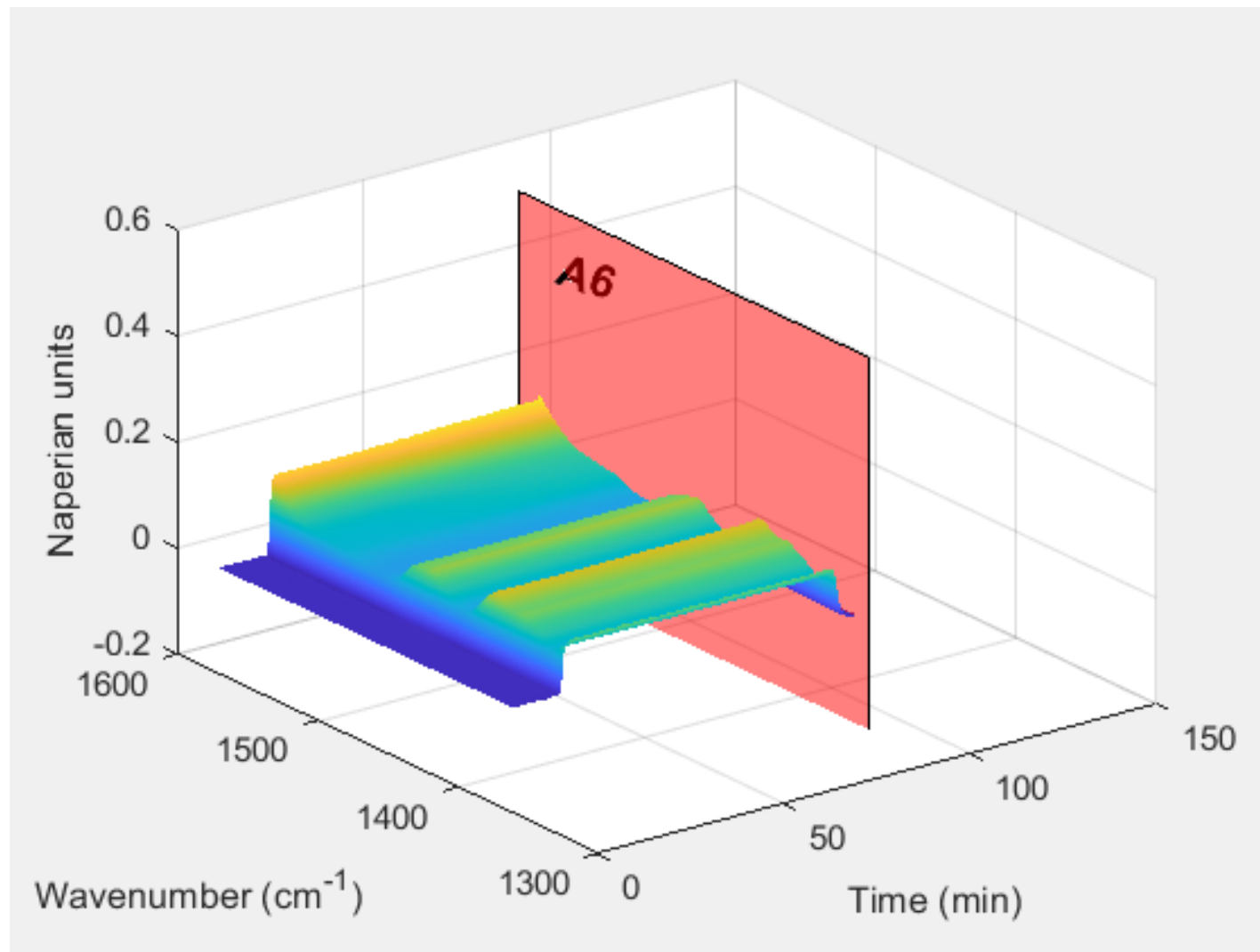
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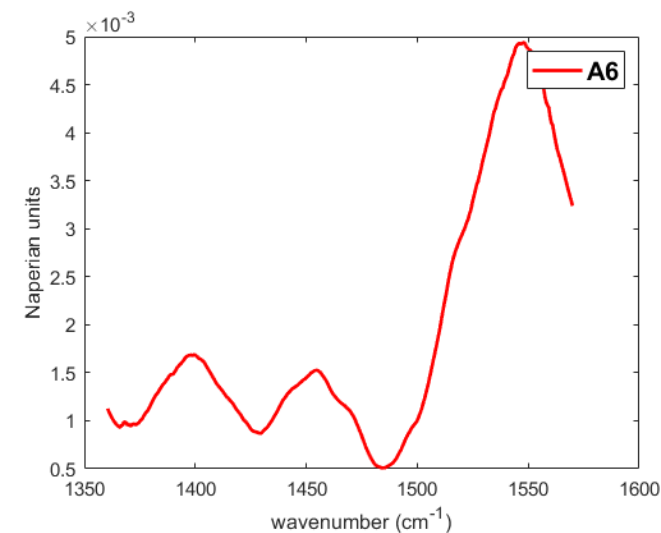
Example #1

Monitoring of a capture step



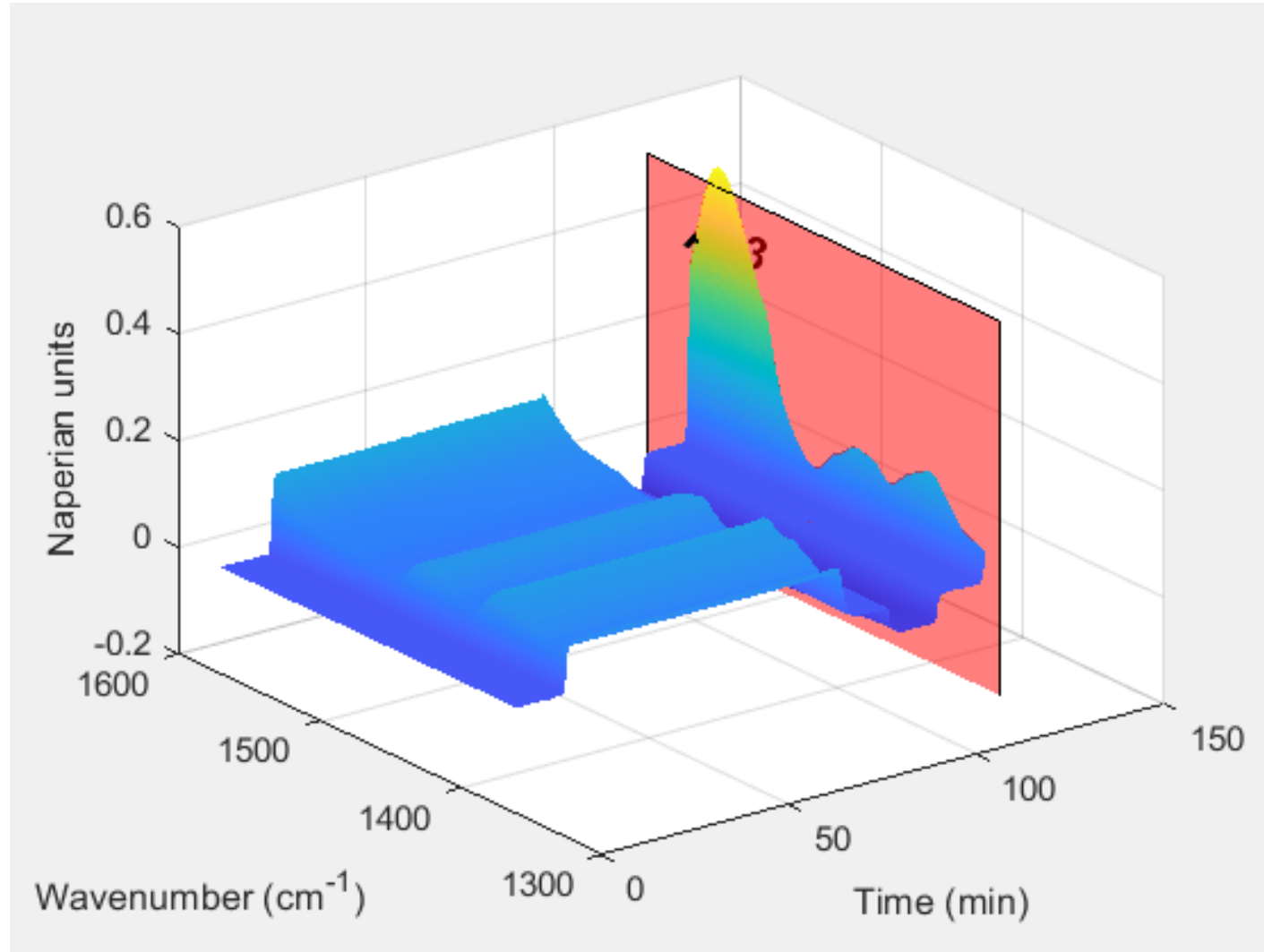
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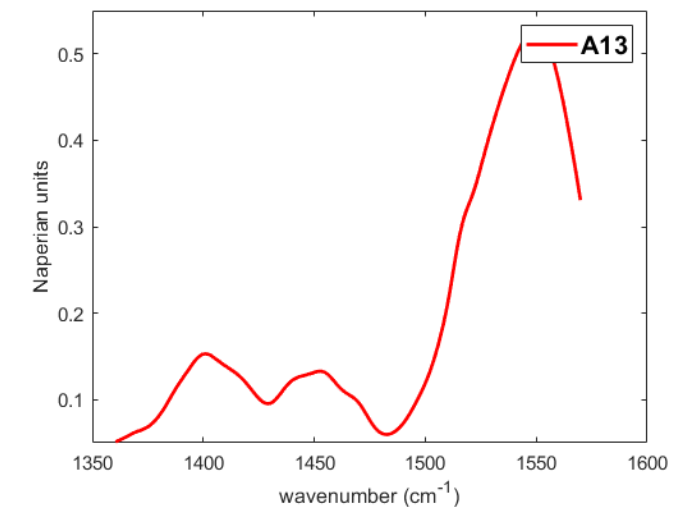
Example #1

Monitoring of a capture step



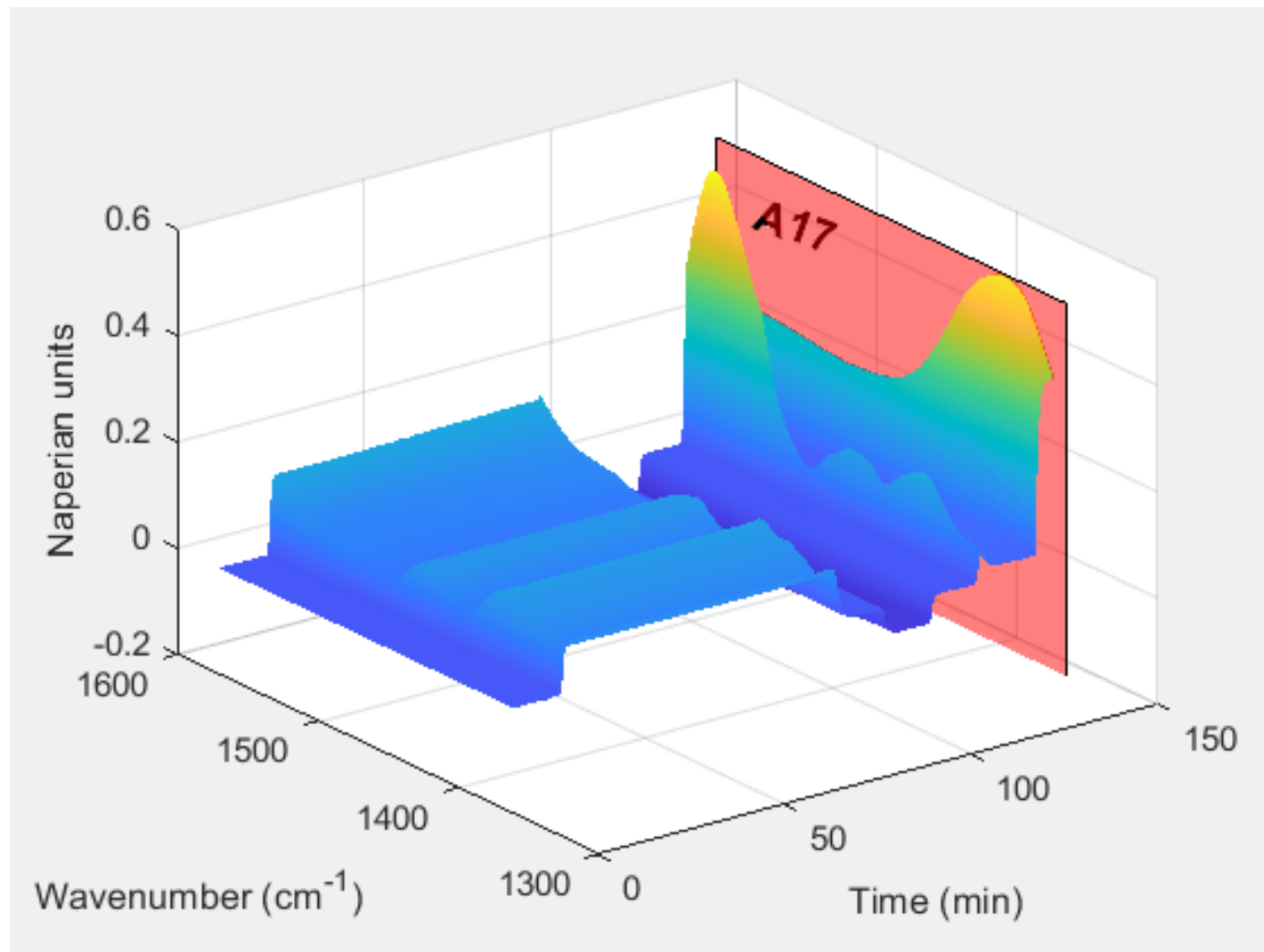
Two-dimensional array of data. At each time point, one can take a slice through the data and extract the characteristic mid-IR spectra that gives a unique fingerprint of the column eluent composition at that time point

The spectra can be extracted for a chosen range of wavenumbers that may be more information reach from a given column eluent composition perspective



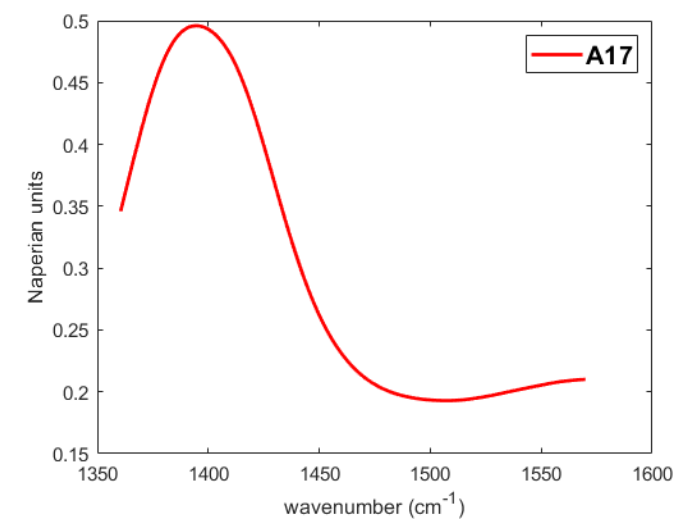
Example #1

Monitoring of a capture step



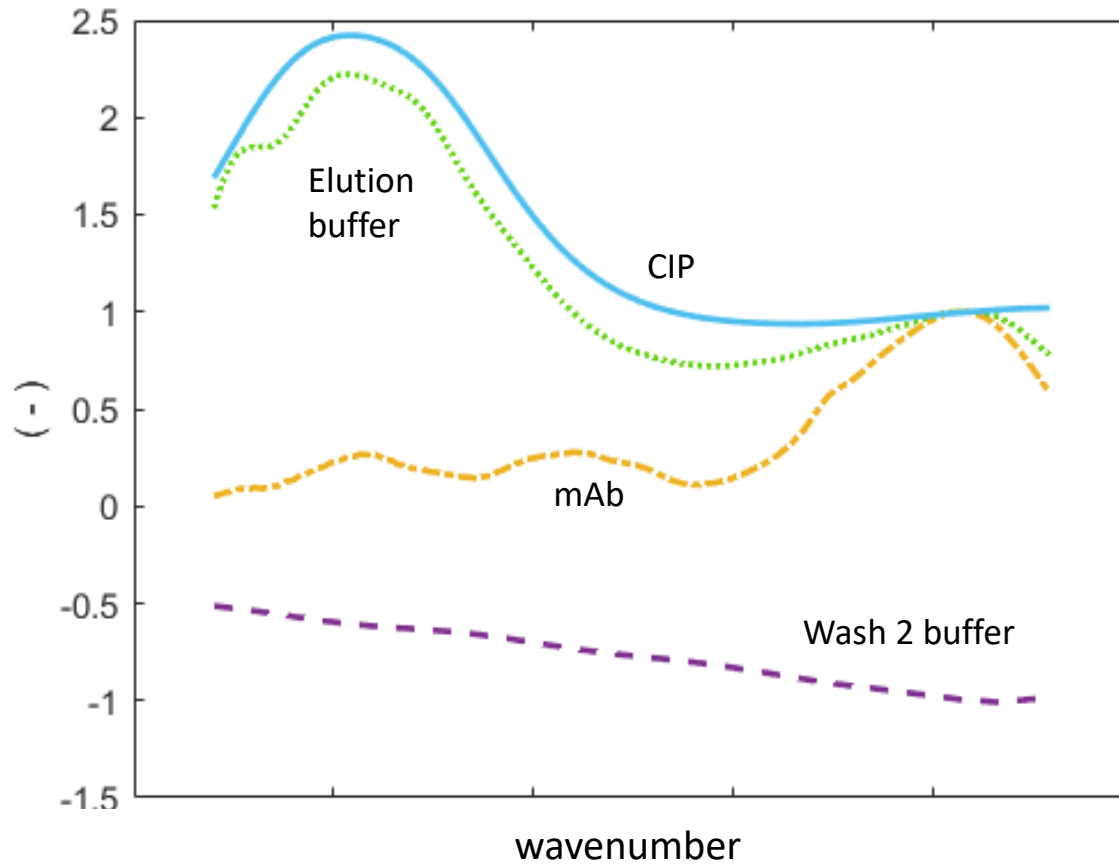
Two-dimensional array of data. At each time point, one can take a slice through the data and extract the characteristic mid-IR spectra that gives a unique fingerprint of the column eluent composition at that time point

The spectra can be extracted for a chosen range of wavenumbers that may be more information reach from a given column eluent composition perspective



Example #1

Monitoring of capture step: mid-IR data deconvolution



Mid-IR spectrum measured at time t is equal to the sum of spectra of individual components multiplied by their concentrations

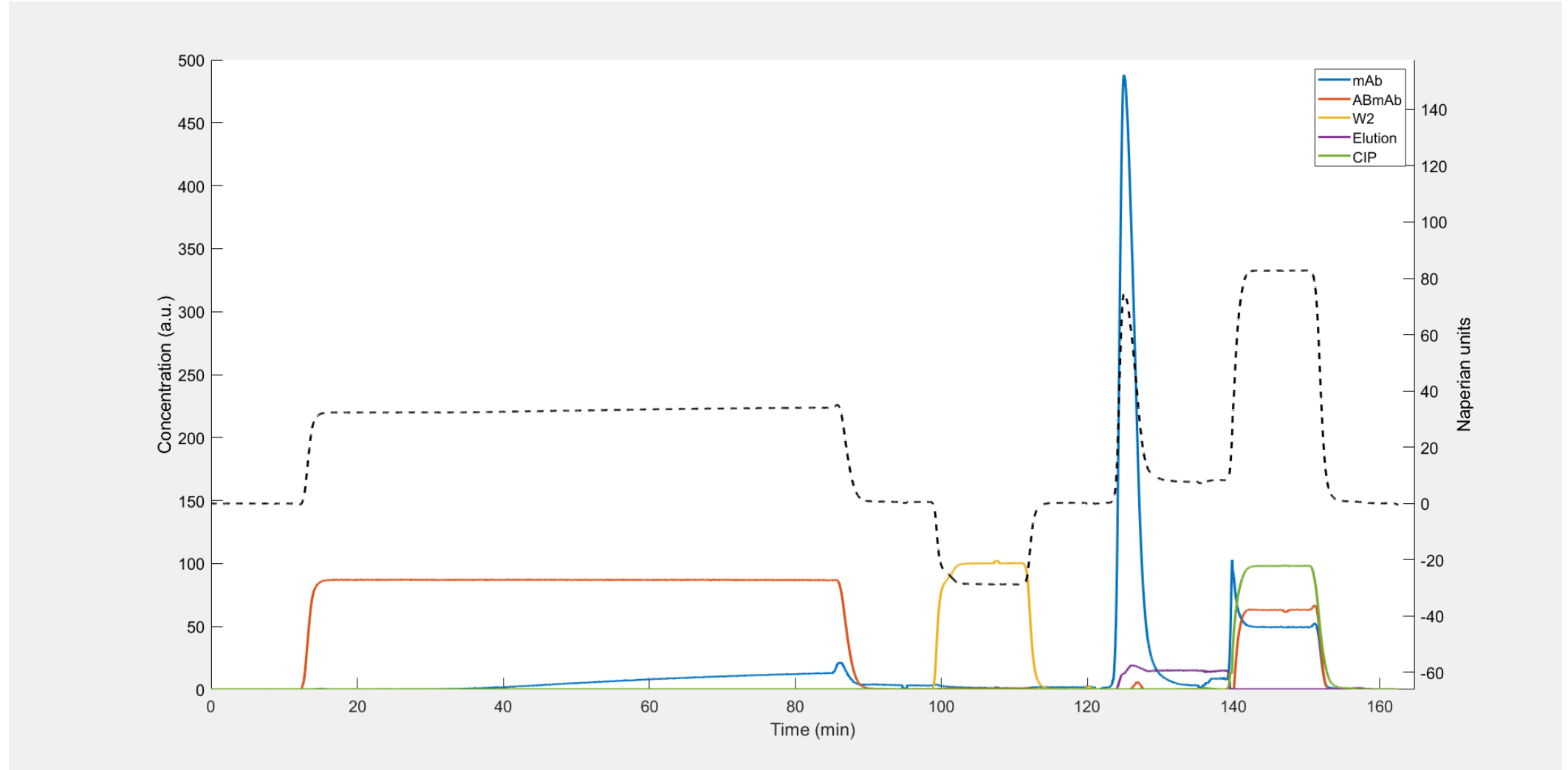
$$\left[\sum_{i=1}^n A_i \right]_t - \left[\sum_{i=1}^n A_{i,est} \right]_t = \left[\sum_{i=1}^n A_i \right]_t - \left[l \sum_{j=1}^{N_c} C_j \sum_{i=1}^n \varepsilon_{i,j} \right]_t = \Delta_t$$

Assuming that experimental noise is small compared to the signal measured, it follows that the residuals Δ_t must be equal to the sum of spectra of all the unknown components, k

$$\Delta_t = \left[l \sum_{k=1}^{N_{unk}} C_k \sum_{i=1}^n \varepsilon_{i,k} \right]_t$$

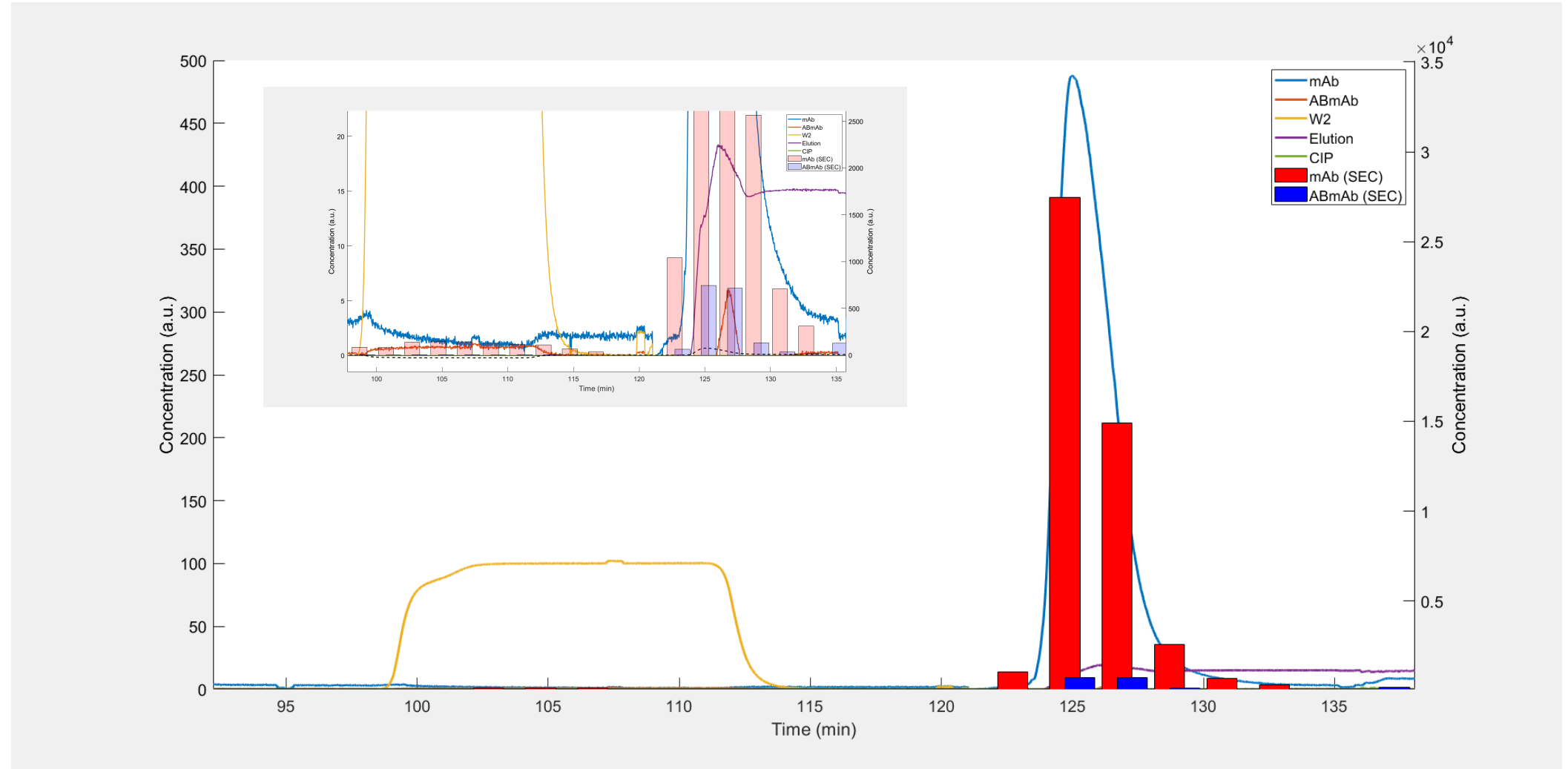
Example #1

Monitoring of capture step: mid-IR derived concentration profiles



Example #1

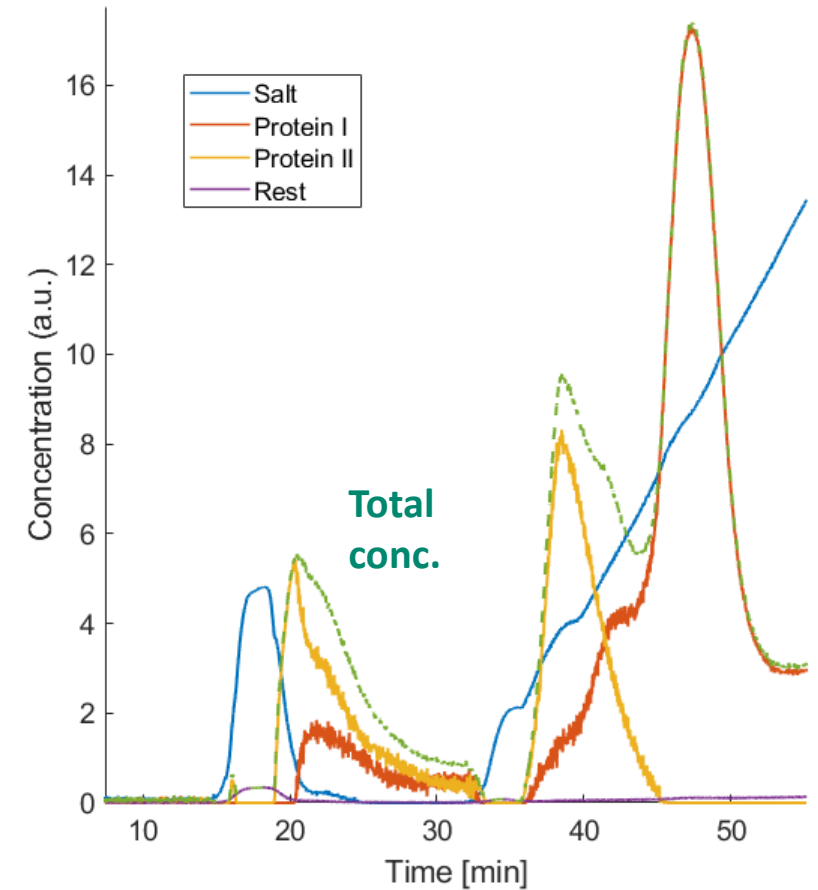
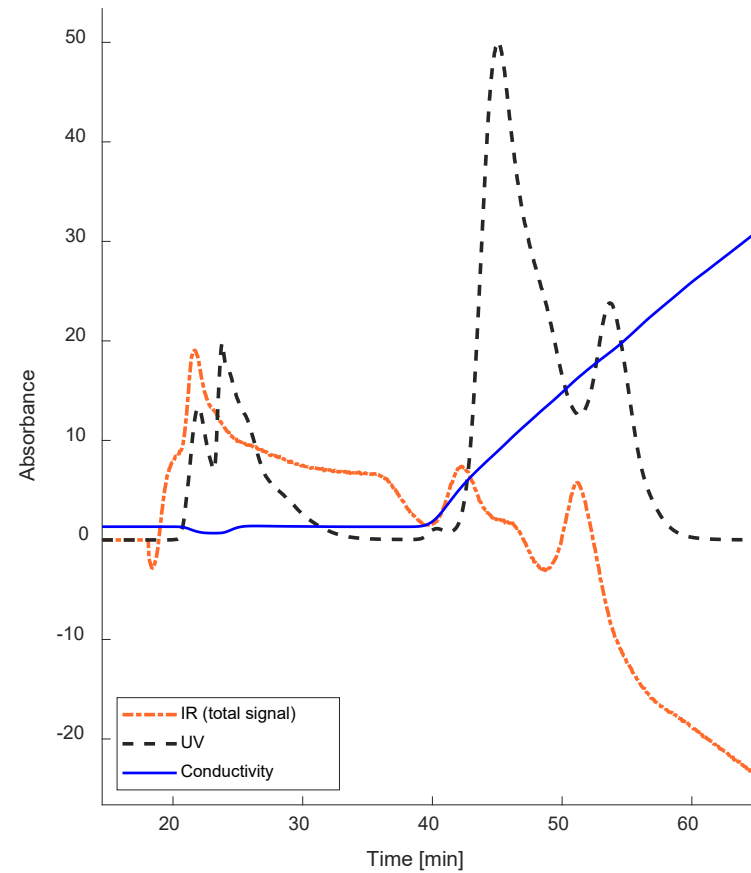
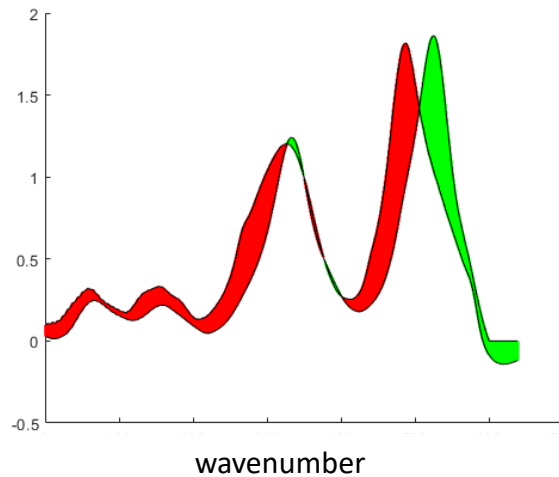
Monitoring of capture step: mid-IR derived concentration profiles



Example #2

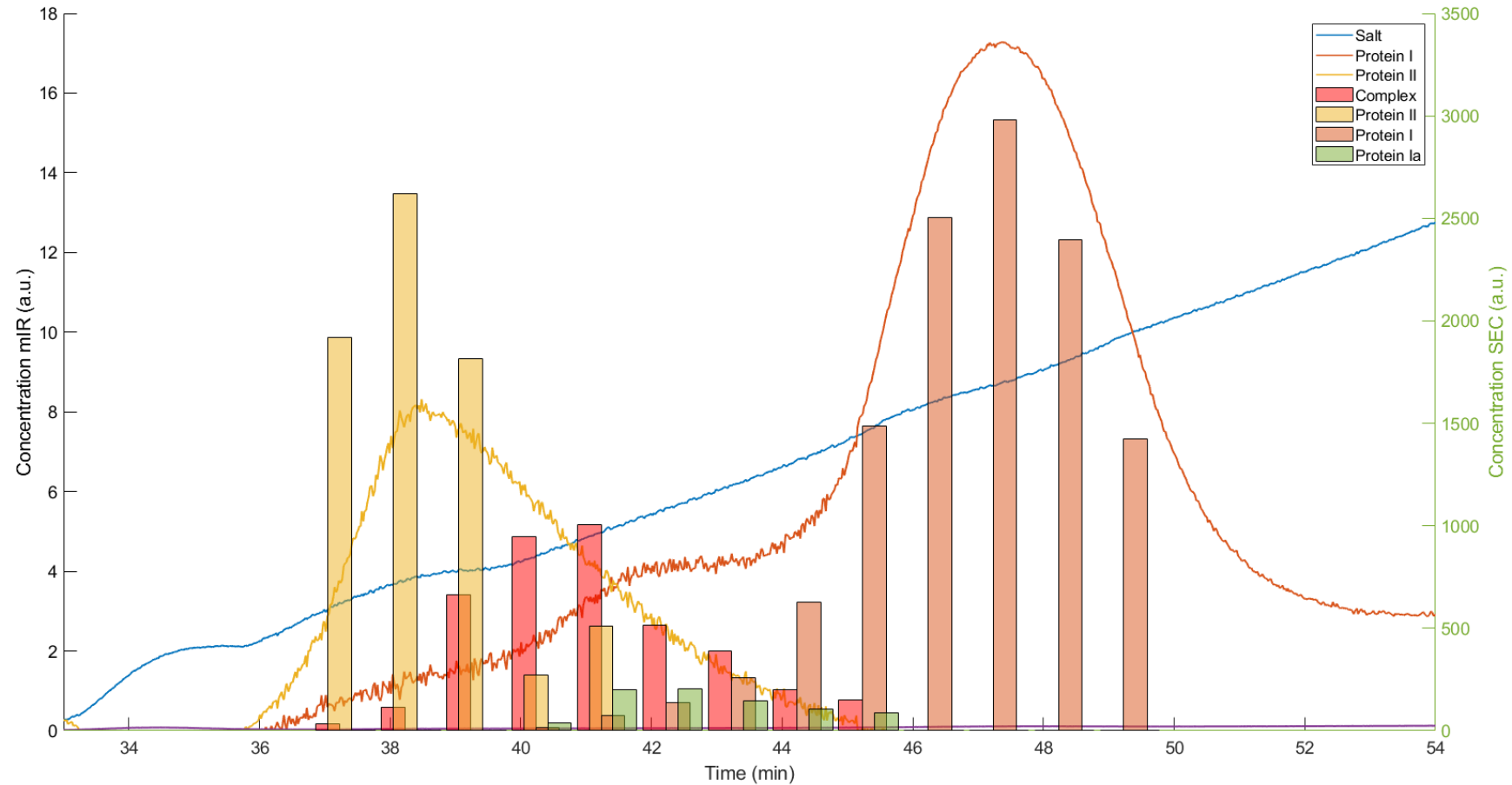
Monitoring of concentration profiles during gradient chromatography

Mixture of Protein I and Protein II
(with a significantly different IR
spectra) was separated using salt
gradient on a cation exchange column



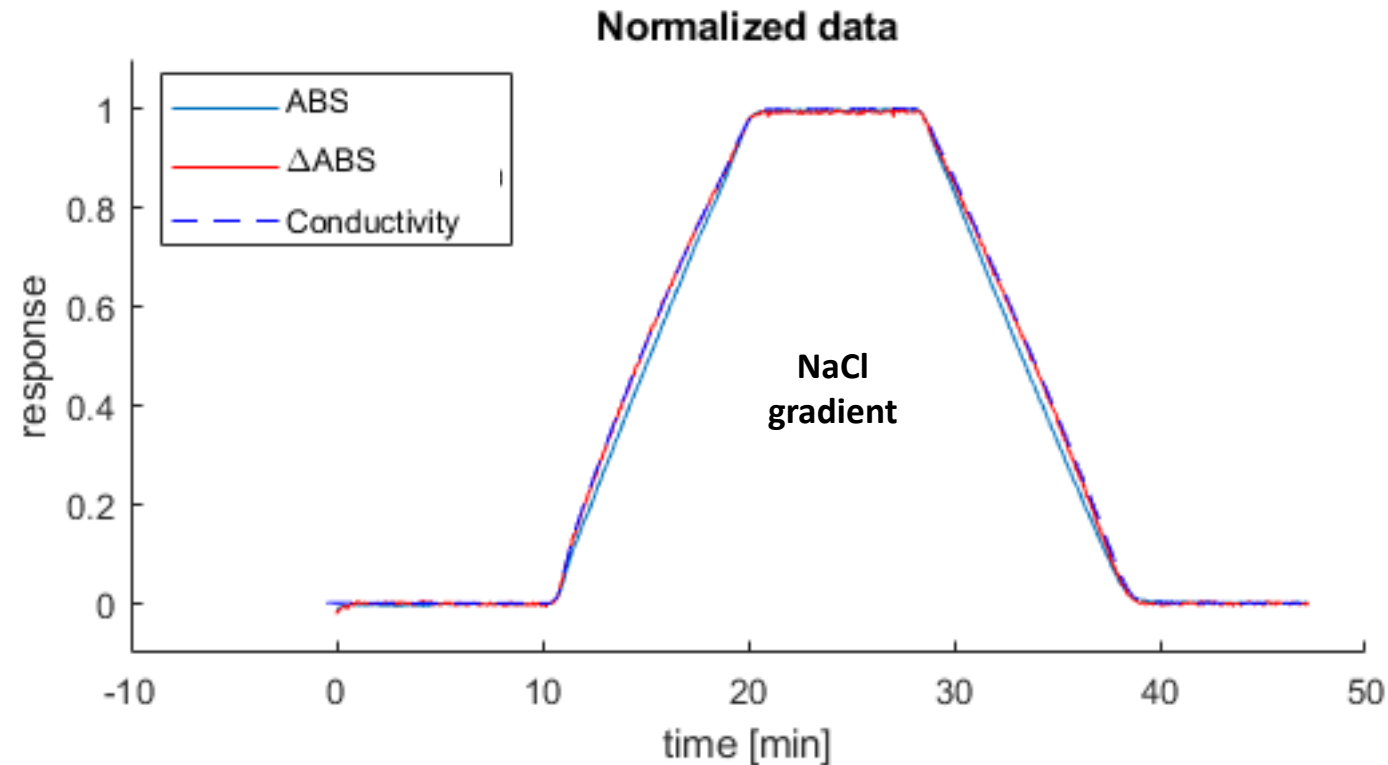
Example #2

Monitoring of concentration profiles during gradient chromatography



Example 3

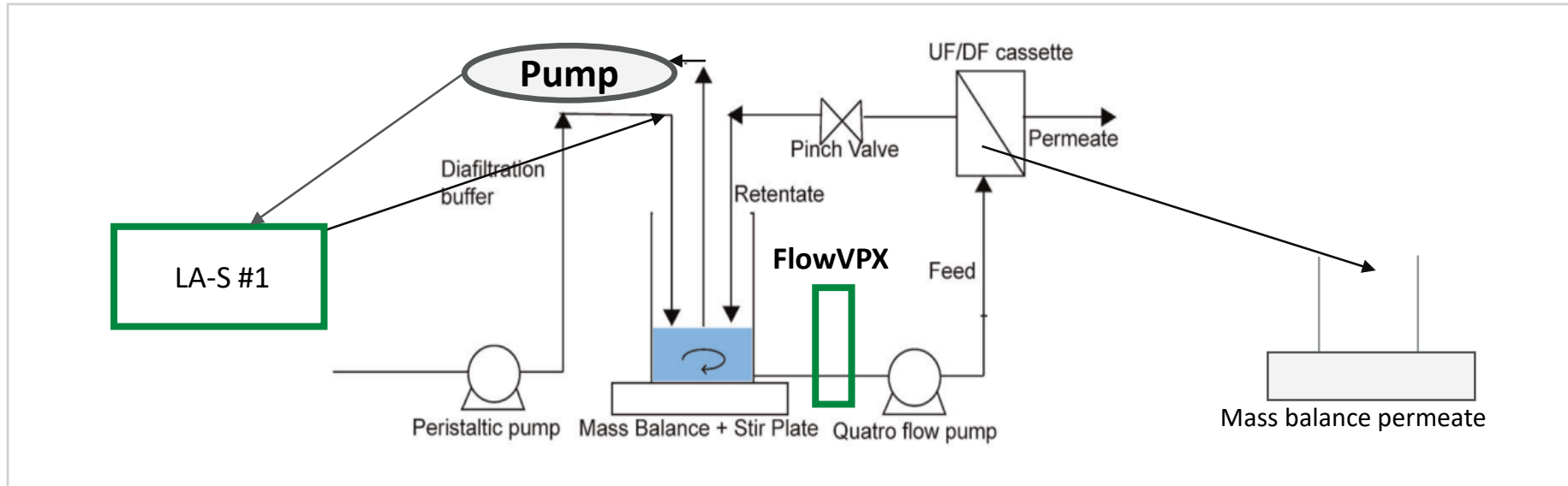
Conductivity meter



- QCL-IR detector can be used to monitor conductivity changes in the presence and absence of proteins
- For certain buffers, *e.g.*, phosphate, QCL-IR detector can be used to monitor changes in pH

Example #4

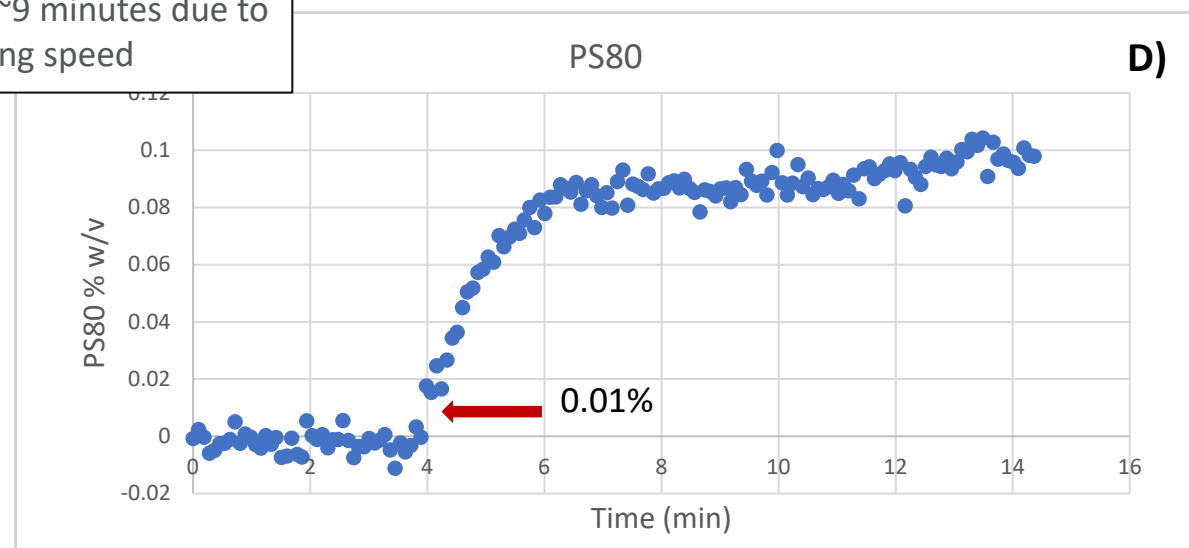
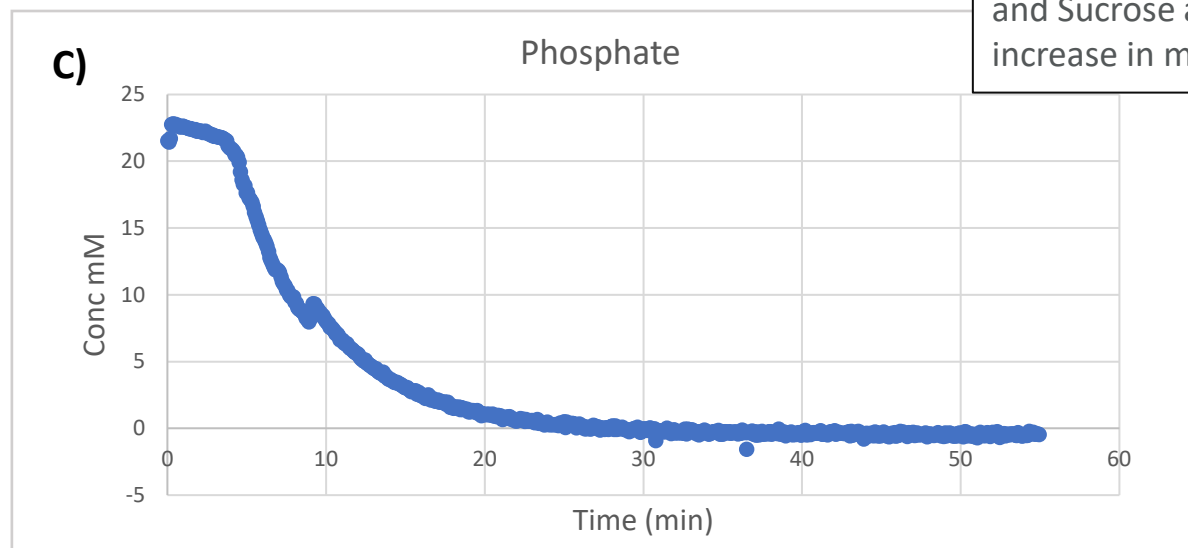
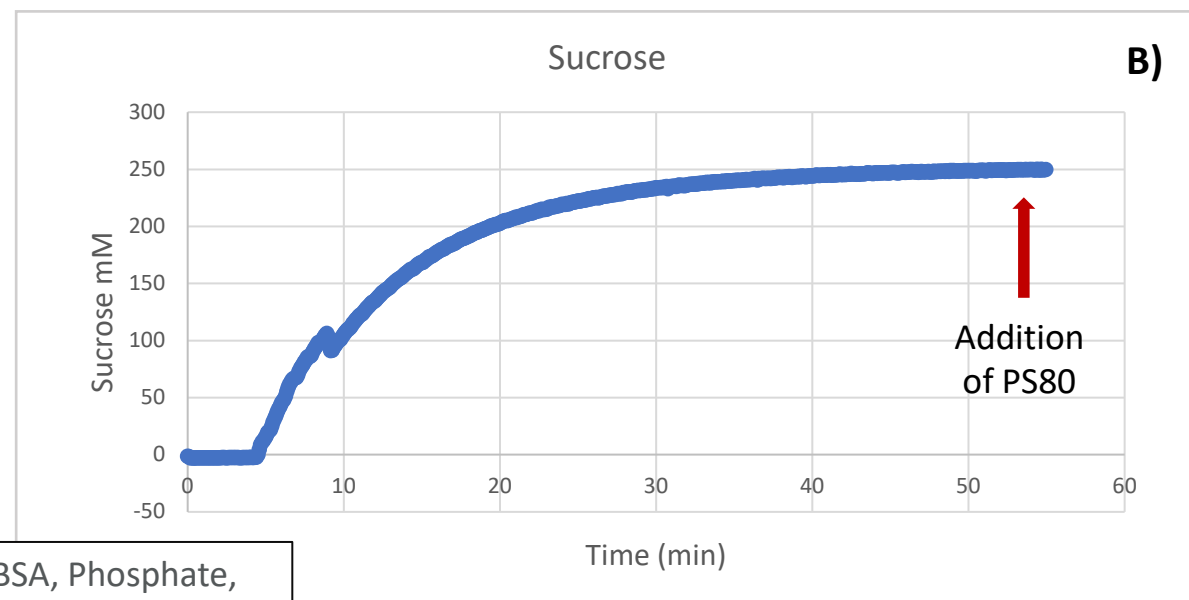
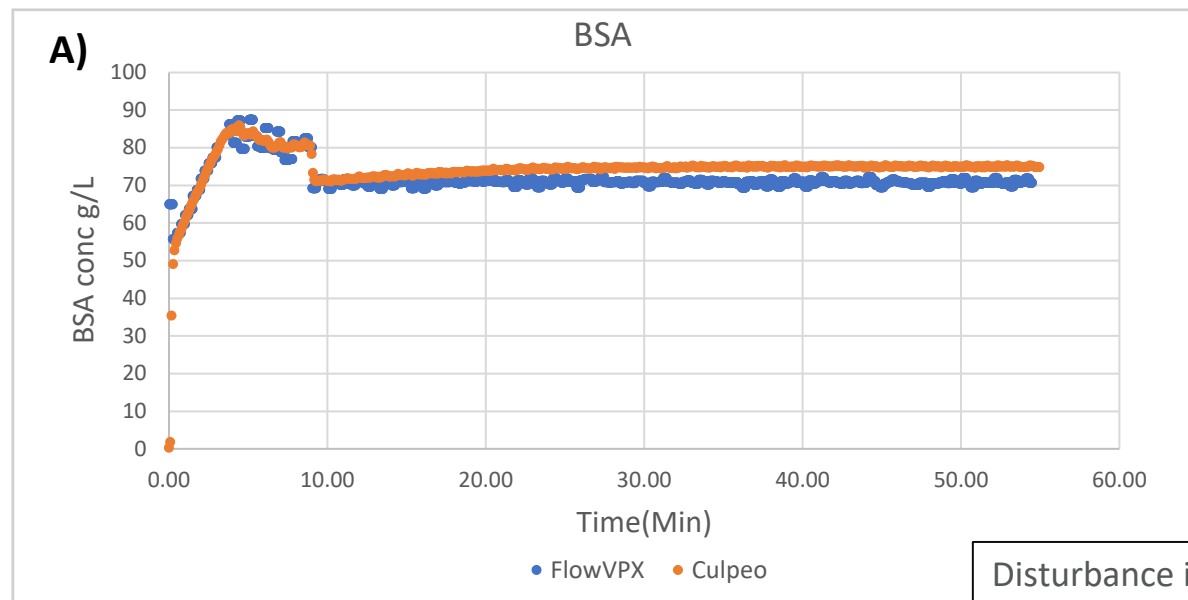
Monitoring of a formulation process



- 0.03 m² total cassette area, 20 psi TPM, 30 LMH Flux, 130mL/min
- Monitor retentate mix with LA-S #1 via HPLC pump recirc at 5mL/min
- Flow Cell: 76um
- BSA (350mL) in 20mM phosphate @ 40 g/L concentrated to 80 g/L
- After reaching UF1 (80 g/L) diafiltered into 30 mM histidine and 250 mM sucrose buffer over 6x DVs
- Monitor phosphate, protein, histidine, and sucrose
- Spike 0.1% PS80, monitor PS80

Source: J.M. West, 2024 ACS Annual Meeting, New Orleans, LA

Example #4



Disturbance in BSA, Phosphate, and Sucrose at ~9 minutes due to increase in mixing speed



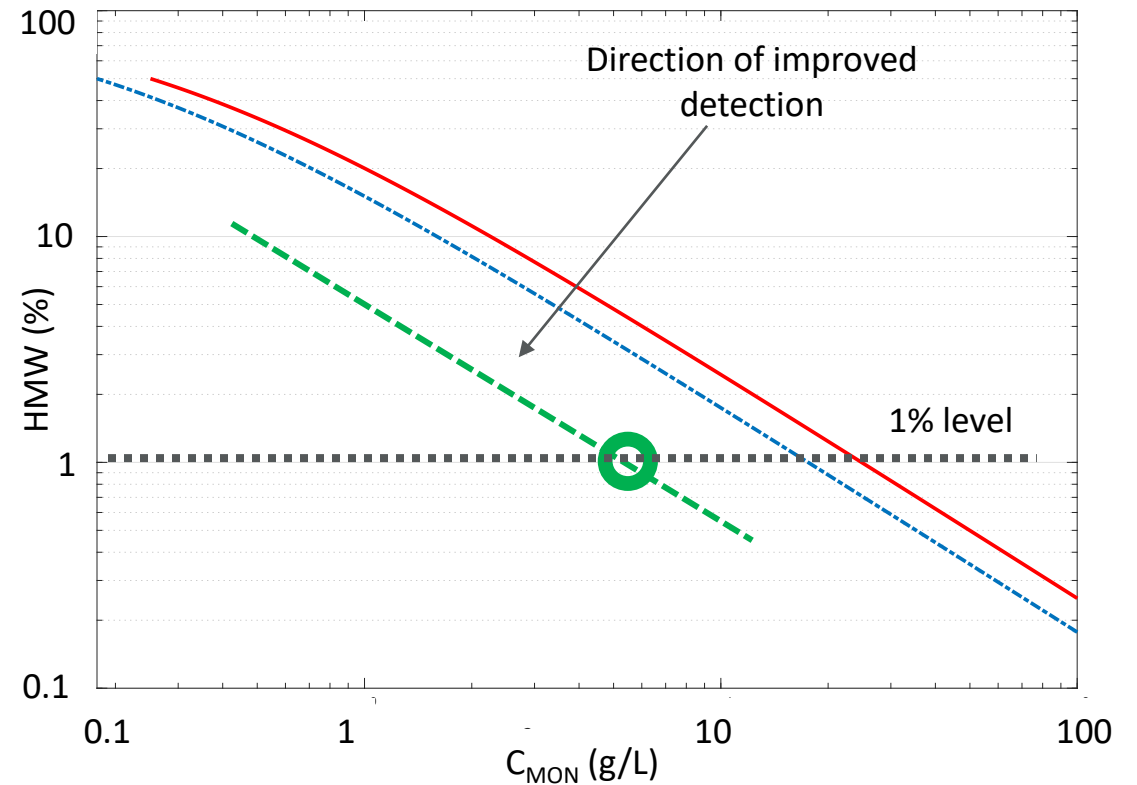
Limit of Detection and other challenges

Current LODs and future specifications

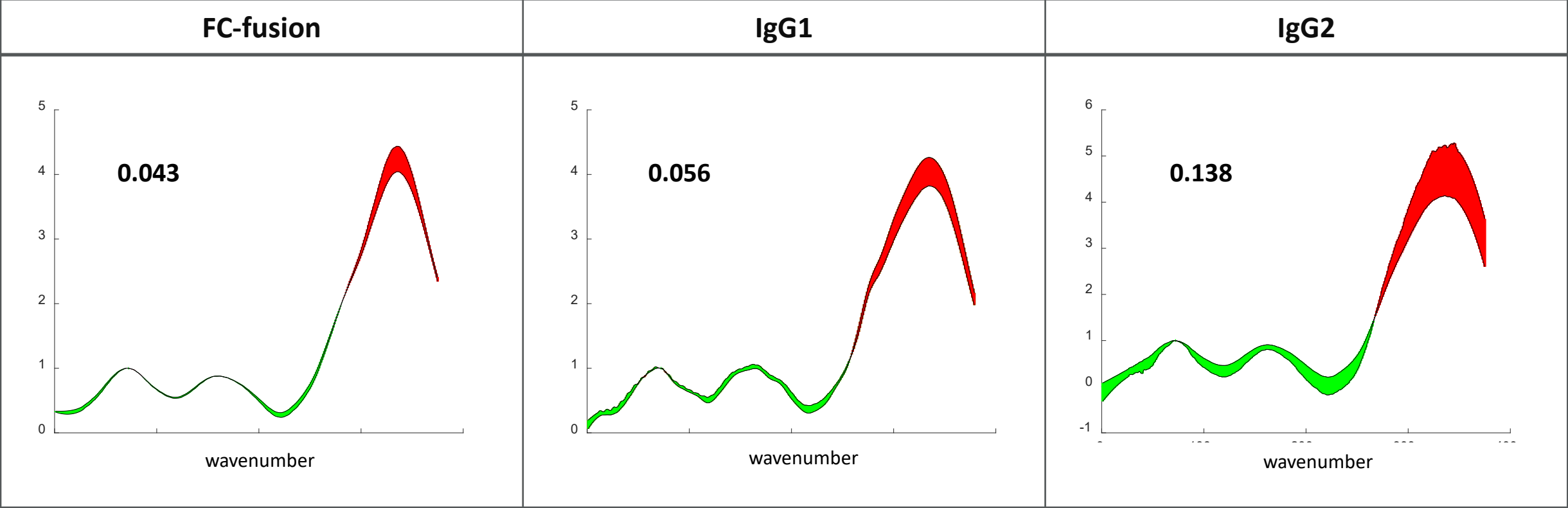
Examples

Type	LOD
Proteins	0.25 g/L
DNA	0.015 g/L
Amino acids	< 7 mM (<i>e.g.</i> 0.8 mM)
Surfactant	0.01%
Carbohydrates	< 1 mM
Buffers	< 1.5 mM
Obtained using either LA-S or LA-P units w/ 51 or 76 um flow cells	

Example detection limits for product related impurity at 1% difference in IR spectra of the product and the impurity



Examples of differences between monomer and aggregates



improved detectability

Conclusions

- mid-IR QCL detector enables very fast (i.e., even real time), and simultaneous, measurements of concentrations of multiple components in a process stream making the detector a very appealing candidate for the ultimate PAT tool
- QCL mid-IR detector provides assurance of quality and/or avoidance of out of spec batches
 - delivers high accuracy and high precision data even at sub second analysis time making it a perfect tool for in-line monitoring of downstream and formulation processes
 - can be used at-line as a quick quality control tool
 - can work as a redundant conductivity meter and, potentially, even as a pH meter
- The QCL based mid-IR instrumentation may have significant advantages over other multi-attribute spectroscopic methods in current use for GMP PAT applications due to ease of data analysis
- mid-IR QCL detector can deliver another level of control when use for monitoring of multicolumn capture step operated in the PCC mode

Acknowledgments

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- Mike Clabough
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- James Peyser
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- George Sherry
- Dodd Weisenberger
- Jay West

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- Ted Stinson
- Edeline Fotheringham

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Thank you



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